

Q.18 A Carnot Cycle machine operates between the temperature limits of 47° and -30° C. Determine the C.O.P. when it operates as:

- a) A Refrigerating machine
- b) A heat pump

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Write a short note on any two:

- a) Explain in brief Carnot cycle of refrigeration
- b) Vapour Absorption refrigeration system
- c) Bell Coleman Cycle of refrigeration

Q.20 Explain the construction and working of an ammonia-water absorption refrigeration system with a neat sketch.

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Level 5, 1st Sem / DVOC (Ref. & Air Cond.)

Subject : Basics of Refrigeration

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

- Q.1 In refrigeration, the application of latent heat is used to:
- a) Cool the condenser
 - b) Absorb heat at low temperature.
 - c) Perform mechanical work
 - d) Compress the refrigerant
- Q.2 The Carnot cycle is considered an ideal refrigeration cycle because:
- a) It uses air as refrigerant
 - b) It has no expansion process
 - c) It operates between two temperature limits with maximum efficiency
 - d) It requires no compressor
- Q.3 Which one of the following is not true for superheating the refrigerant before compression?

- a) It increases the specific volume
 - b) It avoids liquid entering the compressor
 - c) It improves system reliability
 - d) It decreases the compressor work
- Q.4 In a vapour absorption system, the function of the absorber is to:
- a) Evaporate the refrigerant
 - b) Absorb the refrigerant in absorbent
 - c) Provide heat
 - d) Compress the vapour
- Q.5 Which of the following is not a desirable property of a refrigerant?
- a) High toxicity
 - b) Low boiling point
 - c) High latent heat of vaporization
 - d) Low viscosity

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 The unit of refrigeration commonly used in practice is _____ of refrigeration (TR).
- Q.7 The COP of a refrigerator is defined as the ratio of the refrigerating effect to the _____ input.

(2)

188451

- Q.8 Subcooling increases the refrigerating effect by reducing the _____ temperature of the refrigerant.
- Q.9 In the ammonia-water system, _____ acts as the refrigerant.
- Q.10 An ideal refrigerant should have a low boiling point and high _____ heat of vaporization.

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

- Q.11 What are conditions of COP for its maximum value?
- Q.12 Explain various properties of refrigerant : Ammonia and Freon.
- Q.13 Explain reversed carnot cycle refrigeration.
- Q.14 What is sub-cooling of refrigerant and its effects in vapour compression cycle?
- Q.15 Explain primary refrigerants with examples.
- Q.16 Write the principle of vapour absorption system.
- Q.17 Explain in brief various methods of refrigeration with neat sketch.

(3)

188451